

NITIN GOYAL

(617)-319-4514 | nitingoyal3150@gmail.com | <https://github.com/nitin3150> | <https://www.linkedin.com/in/nitin-k-g/>

SKILLS

AI / ML: LangGraph, LangChain, RAG Pipelines, LLM Evals, Guardrails, DSPy, LangSmith, LiteLLM, Langfuse, RAGAS, PyTorch, Prompt Engineering, Adversarial Testing

Backend / Infra: FastAPI, Node.js, REST APIs, WebSockets, Docker, Kubernetes, AWS (ECS, Lambda, S3), CI/CD, Git

Databases: PostgreSQL, MySQL, MongoDB, Redis, Pinecone, ChromaDB

Programming Languages: Python, JavaScript, TypeScript, SQL

PROFESSIONAL EXPERIENCE

Founding AI Engineer

May 2026 - Present

AgentSentinel | AI Agent Governance & Safety Platform [[Link](#)]

Remote

- Engineered a **3 stage** agent pre-deployment pipeline (Inspect → Stress-Test → Optimize) using **LangGraph** orchestration and **DSPy** prompt optimization, improving agent pass rate from **60% to 85%** across adversarial test runs.
- Built a stress-testing engine that fires **100 adversarial prompts** per run against live agents across **4 compliance rule sets (HIPAA/SOC2/OWASP)**, surfacing jailbreak, data leakage, and misleading outputs before production deployment.
- Designed a **provider-agnostic** eval architecture via **LiteLLM**, enabling zero-code-change pipeline execution across **100+ LLM providers**, making enterprise-grade agent safety accessible without infrastructure changes.

AI Engineer

Jul 2025 - Sep 2025

Cognitive Health Technologies

New York, NY

- Built a **multi-agent LangGraph** pipeline for end-to-end healthcare claims processing, integrating **OCR** and **LLM** models via **REST APIs**, improved claim accuracy by **80%**.
- Designed **feedback-driven** document review loops with **human-in-the-loop** verification, cutting manual review time from **48h to 6h** while maintaining full data quality.
- Implemented **RAGAS + LLM-as-a-Judge** evaluation pipelines that reduced hallucination rates by **40%**, enabling **continuous model monitoring** across production healthcare workflows.
- Engineered **secure API** data handling with **role-based access** controls and immutable **audit logging**, ensuring **full PHI traceability** and **HIPAA** compliance across all agent interactions.

Software Developer

Mar 2021 - Jul 2024

Open Access Technologies International

India

AMIOT - Energy Forecasting & Anomaly Detection

Sep 2021 - Jul 2024

- Productionized **XGBoost + LSTM** ensemble models processing **1M+** smart meter records/day, improving energy grid forecasting accuracy by **20%** via **Airflow-orchestrated** automated retraining.
- Deployed **isolation forest + autoencoder** anomaly detection with **Kafka** streaming and **Redis-backed** inference, reducing false-positive alerts by **35%** across production workloads.
- Designed and maintained **ETL pipelines** ingesting daily sensor data with automated validation, cutting model deployment and retraining cycles by **30%**.

WebOMS - Web Operations & Monitoring Dashboards

Mar 2021 - Sep 2021

- Refactored legacy **SQL queries** with indexing optimizations, reducing backend response times by **30%**.
- Integrated **realtime IoT sensor** streams into predictive maintenance models, enabling automated retraining triggers on anomaly detection thresholds across peak load periods.

PROJECTS

Saddle | Self-Improving Multi-Agent Orchestration Harness [[Link](#)]

June 2026

Best Use of Weave · Multi-Agent Orchestration Hackathon · Python, LangGraph, DSPy, W&B Weave

- Built a self-improving multi-agent harness using LangGraph that autonomously diagnoses coordination failures and rewires team topology, lifting correctness **65% to 92%** and multi-hop retrieval accuracy **36% to 93%**.
- Engineered **5 Weave-instrumented** coordination scorers feeding a one-change-per-generation optimizer with automatic regression detection, enabling exact attribution of every performance delta across **26 benchmark questions**.

EDUCATION

Northeastern University

Boston, MA

Masters in Artificial Intelligence | GPA: 3.95/4.0

Relevant Coursework: AI in Financial Services, NLP, Time Series Analysis, Deep Learning, BigData

Punjabi University

India

B.Tech in Computer Science Engineering | GPA: 4.0/4.0

Relevant Coursework: Data Structures, Data Mining, Design and Analysis of Algorithm, Object Oriented Programming